

Problem 6

Problem 6

Problem 1 Consider a Riemann sum with n rectangles for the function f on the interval $[a, b]$.

Use geometry to find the limit of Riemann sums as $n \rightarrow \infty$.

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x \text{ ?}$$

(a) $f(x) = x, [0, 3];$

(b) $f(x) = |x|, [-3, 3];$